
Additional Problems for Ch 15 HRW – Oscillations
 Addisionele Probleme vir Hfst 15 HRW – Ossilasies

	9 th Edition	8 th Edition
Problems	14, 23, 35, 40, 41, 49	22, 17, 33, 44, 43, 53

14	22	$\omega = 7.07 \text{ rad/s}$. Use Energy methods (discussed in §15-4) or trigonometric techniques based on Eq. 15-3 and Eq. 15-6 to solve the problem. (a) $x_m = 0.500 \text{ m}$. (b) $\phi = -8.38 \text{ rad}$ $x_o = x_m \cos \phi = -0.251 \text{ m}$. (c) $v_o = -x_m \omega \sin \phi = 3.06 \text{ m/s}$.	
23	17	$x_m = 0.028 \text{ m}$	$x_m = 0.031 \text{ m}$.
35	33	$\omega = 1800 \text{ rad/s}$. (a) $T = 2\pi/\omega = 3.5 \times 10^{-3} \text{ s}$. (b) $v_m = 3.6 \text{ m/s}$. (c) The total energy is $\frac{1}{2} kx_m^2 = \frac{1}{2} mv_m^2 = 0.080 \text{ J}$. (d) $F = 64 \text{ N}$ (e) $F = 32 \text{ N}$	$\omega = 2000 \text{ rad/s}$. (a) $T = 2\pi/\omega = 3.1 \times 10^{-3} \text{ s}$. (b) $v_m = 4.0 \text{ m/s}$. (c) The total energy is $\frac{1}{2} kx_m^2 = \frac{1}{2} mv_m^2 = 0.080 \text{ J}$. (d) $F = 80 \text{ N}$ (e) $F = 40 \text{ N}$
40	44	$d = 0.056 \text{ m}$.	
41	43	(a) $I = 0.203 \text{ kg} \cdot \text{m}^2$. (b) $d = 0.483 \text{ m}$ (c) $T = 1.5 \text{ s}$	(a) $I = 0.205 \text{ kg} \cdot \text{m}^2$. (b) $d = 0.477 \text{ m}$ (c) $T = 1.5 \text{ s}$
49	53	(a) $\phi = 0.845 \text{ rad}$ (b) $\theta_m = 0.0602 \text{ rad}$.	

